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PROJECT

Eagles

[app.eagles-eg.online/](#) • July 2026 - Present

- Engineered a production workspace-rental management platform using **React.js**, automating lead ingestion, workload-based assignment, lifecycle tracking, and lead-to-customer conversion to replace manual spreadsheet workflows.
- Architected a deterministic lead-assignment engine that balances workload across eligible sales representatives across 4 open-lead statuses, using Round Robin as a tie-breaker and enforcing a 30-minute acknowledgment SLA with automated escalation and reassignment.
- Integrated 2 lead-ingestion channels — Meta Lead Ads webhooks with HMAC-SHA256 signature verification and WhatsApp via API — and implemented normalized mobile-number identity matching and message-level delivery idempotency.
- Enforced server-side row-level access control across 3 application roles (Admin, Manager, Employee), restricting Employee visibility to assigned leads only and validating the boundary with a 14/14 authorization regression suite.
- Deployed and maintained a self-hosted production stack combining Docker Compose, Pigsty-managed PostgreSQL, Cloudflare Tunnel, and pgBackRest with continuous WAL archiving and hourly off-site differential backups to Cloudflare R2.
- Delivered a role-aware CRM dashboard featuring SLA countdown tracking, urgency-grouped notifications, lead-velocity and conversion-funnel analytics, team performance rankings, and a 7-stage lead-pipeline kanban board.

Visa Appointment Automation

August 2026 - August 2026

- Built a single-POST HTTP scanner that bypasses the appointment portal's full multi-step wizard, averaging ~270ms per request and dispatching up to 24 parallel requests per scan cycle across an 8-week rolling horizon for up to 20 active client jobs.
- Developed Durable Object-backed distributed locks with atomic acquire, seal, and release states, ensuring zero double bookings during concurrent job triggers, validated through custom concurrency testing.
- Automated the full booking wizard using Cloudflare's Playwright binding, completing a six-step, 31-field form and confirming success only by parsing the portal's GESX reference from the server response, not by assuming UI completion.
- Diagnosed a 0-of-3 success rate for image-based CAPTCHA solving (Tesseract OCR and Vision LLM) and redesigned the pipeline around the portal's audio channel, transcribing BotDetect sound challenges with two Whisper models run concurrently.
- Eliminated a paid third-party CAPTCHA-solving API by replacing it with Cloudflare Workers AI and open-source Tesseract.js, keeping the system within Cloudflare's target through prompt-layer controls — 15-token response caps and structured 4–6 character output parsing — rather than model fine-tuning.
- Hardened PII storage with AES-256-GCM encryption via Web Crypto and PBKDF2 at 100,000 iterations, cached in memory for ~87x faster bulk decryption, backed by dual-layer authentication and session cookies that store only a SHA-256 hash of the admin key.

AI Skills Aggregator

[github.com/abdulsamed1/AI-skills-bank](#) • May 2026 - Present

- Built a hybrid keyword-and-LLM classification pipeline in **Rust** for an 8,000+ file skill catalog, using strict schema-constrained prompts, concurrent batching, and jittered exponential backoff with circuit-breaking for deprecated models.
- Reduced redundant LLM calls via keyword pre-filtering and SHA-256-based classification caching, tracked with atomic hit/miss/error counters for real-time cache efficiency monitoring.
- Designed a pluggable LLM provider interface and a custom proxy adapter, with per-provider endpoint, credential, and model resolution, extending effective coverage to 12+ free-tier providers.
- Mitigated malformed LLM outputs with a resilient parser that strips Markdown fences, recursively recovers nested or malformed JSON, tolerates model-introduced field-name typos, and defaults missing confidence scores without causing hard failures.

Intellichain

[github.com/abdulsamed1/Intellichain](#) • March 2025 - April 2025

- Architected a 4-component Tweet-to-NFT claim pipeline that integrates an ElizaOS AI agent, Chainlink Functions, Supabase, and an ERC-721 smart contract to parse tweet replies for wallet addresses, claim codes, and claim intent.
- Engineered a 2-stage, 75-line Docker build pipeline that provisions Foundry, Slither, Cyfrin tooling, and Aderyn, automating dependency installation, optimizing Solidity builds, and executing static analysis.
- Developed and deployed a 143-line ERC-721 GetGift contract on Sepolia (June 2025) with Chainlink Functions callbacks, allow-list access control, reentrancy protection, and 3 predefined NFT reward types backed by IPFS metadata.
- Implemented Chainlink Functions validation with a 300,000-gas callback budget, querying Supabase for claim-code validity and enforcing single-use redemption on-chain before minting the NFT to the verified wallet.
- Managed 3 Solidity dependencies—OpenZeppelin Contracts, Chainlink, and forge-std—as Git submodules and automated recursive installation and builds through Foundry and Makefile workflows for reproducible compilation.

Ride Hailing Aggregator SaaS

[opran.org/](#) • August 2025 - July 2026

- Built a tenant-isolated, API-key-authenticated WebSocket pricing gateway on Cloudflare Workers, enforcing per-tenant rate limits and 120-second TTL-based connection leases using Durable Objects.
- Consolidated 27 functional requirements across 7 platform areas, including API execution, billing, rate limiting, and monitoring, behind a single WebSocket endpoint secured by a unified API key.
- Secured API access with five authentication and authorization layers, including timing-safe SHA-256 API key validation, HMAC-signed, session-bound CSRF tokens, and instant KV-backed session revocation.
- Hardened asynchronous billing and credential-refresh workflows with a cursor-based batch scheduler, atomic D1 transactions, and automated webhook dead-letter-queue reconciliation.
- Verified dual-provider billing integrations with Polar through constant-time webhook signature checks and 129 passing unit tests spanning activation, cancellation, and refund flows.
- Authored 16 architecture decision records and 11 operational runbooks covering D1 migration governance, incident response, disaster recovery, and rollback procedures.

Freellmapi

Open-source Contribution • [github.com/abdulsamed1/freellmapi](#) • July 2026 - July 2026

- Architected an SQLite-backed conversation memory that combines FTSS full-text search with recursive-CTE session lineage reconstruction.
- Engineered WAL concurrency controls (1-second busy timeout, jittered retries, checkpoints every 50 writes) and verified reliability across a 206-test suite spanning 27 files with zero build errors.
- Implemented a native MemOS memory provider that translates OpenAI-compatible messages into MemOS queries, enforces an 8-turn/12,000-character context cap with token authentication and cross-format response normalization, and registers it across 3 LLM models with migration and fallback support.
- Built OpenAI-compatible API support that translates Responses-format input into the internal message schema and converts completions and streaming output into Responses/SSE events, with suffix-based model resolution and tool-call payload normalization.
- Developed a **TypeScript** diagnostics utility that performs live provider-key validation and latency checks, complemented by documentation covering system architecture, data model, and 11 provider integration guides.

EDUCATION

Bachelor's Degree in Management Information Systems

New Cairo Academy • Cairo, Egypt • 2026

SKILLS

